

GENGMA, China

WMO: 569460

Lat: 23.5273N

Lon: 99.3881E

Elev: 3622

StdP: 12.87

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	39.7	41.6	33.4	31.9	69.0	35.9	35.4	63.5	11.3	72.5	9.6	72.5	0.7	110	0.232

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	12.4	89.7	66.8	88.0	67.3	86.4	67.5	73.6	82.2	73.0	80.8	72.6	80.0	5.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.6	133.7	75.8	71.0	131.0	75.4	70.6	129.1	75.1	40.0	82.6	39.5	81.6	38.9	80.5	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.5	7.5	6.0	35.5	93.1	2.0	1.5	34.1	94.1	32.9	95.0	31.8	95.8	30.4	96.8		
			34.8	75.1	1.8	1.0	33.5	75.8	32.4	76.4	31.4	77.0	30.1	77.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.4	57.6	61.5	67.5	72.3	75.4	76.3	75.6	76.1	75.1	71.3	64.5	58.7
	DBStd	7.48	2.50	3.23	3.34	3.43	3.71	2.76	2.64	2.33	2.63	3.35	3.60	3.44
	HDD50	1	0	0	0	0	0	0	0	0	0	0	0	1
	HDD65	602	228	104	15	2	1	0	0	0	0	2	52	198
	CDD50	7067	237	323	542	669	786	788	795	809	754	660	435	269
	CDD65	2193	0	6	92	220	322	338	330	344	305	197	37	2
	CDH74	14077	27	274	1146	1996	2437	1951	1575	1827	1613	961	247	23
	CDH80	3685	0	19	264	695	864	498	325	437	395	177	11	0
Wind	WSAvg	2.1	1.8	2.4	2.8	2.9	2.5	2.2	1.9	2.1	2.0	1.7	1.5	1.4
Precipitation	PrecAvg	52.7	0.8	0.6	0.8	2.1	5.7	8.0	11.6	9.1	6.3	5.1	1.9	0.6
	PrecMax	70.6	7.4	2.2	3.1	8.1	10.7	13.1	16.7	12.9	11.2	13.1	7.0	4.0
	PrecMin	39.1	0.0	0.0	0.0	0.2	1.4	3.9	6.1	5.1	3.0	1.4	0.0	0.1
	PrecStd	8.3	1.4	0.7	0.8	1.7	2.8	1.9	2.8	1.9	2.2	3.0	1.9	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.7	82.4	87.1	91.1	92.6	91.7	89.0	89.0	88.1	86.2	81.6	76.5
		MCWB	56.3	57.6	59.7	62.9	66.5	69.3	72.4	72.8	71.1	68.9	64.3	59.5
	2%	DB	74.4	79.8	85.0	88.8	89.9	87.9	86.1	86.6	86.3	83.9	79.0	74.2
		MCWB	55.4	57.2	59.4	62.7	66.2	70.2	72.3	72.4	71.0	68.1	62.9	58.5
	5%	DB	72.4	77.5	83.1	86.9	87.6	85.2	83.6	84.7	84.4	82.0	77.0	72.2
		MCWB	55.2	56.5	59.1	62.5	66.4	70.5	72.1	72.3	70.8	67.3	62.2	57.7
	10%	DB	70.2	75.2	80.8	84.6	85.1	82.8	81.3	82.4	82.3	79.8	74.7	69.9
		MCWB	54.6	55.7	58.6	62.4	66.7	70.7	71.9	72.0	70.5	67.1	61.5	57.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.9	61.1	63.2	68.0	71.9	73.8	74.3	74.7	73.6	71.5	68.2	63.7
		MCDB	69.6	75.8	77.3	80.2	79.7	82.0	83.2	84.5	83.0	81.2	74.6	69.3
	2%	WB	58.0	59.3	61.9	66.4	70.9	72.9	73.5	73.7	72.7	70.4	66.6	61.7
		MCDB	69.0	74.5	77.1	78.6	78.5	80.4	81.6	82.6	81.4	79.2	72.9	68.7
	5%	WB	56.6	58.0	60.9	65.3	70.0	72.5	72.8	73.0	72.0	69.6	65.3	59.9
		MCDB	68.1	72.9	76.8	78.1	77.5	79.5	80.1	81.1	80.3	77.6	71.5	68.1
	10%	WB	55.4	56.8	60.0	64.4	69.3	71.9	72.3	72.4	71.4	68.8	63.9	58.3
		MCDB	67.0	71.3	75.8	77.8	77.1	78.4	78.8	79.8	79.2	75.9	70.4	67.2
Mean Daily Temperature Range	5% DB	MDBR	28.9	29.2	27.1	23.5	17.3	12.4	11.0	13.0	15.1	17.4	22.6	25.9
		MCDBR	32.2	32.7	30.6	27.2	22.4	17.5	15.5	17.2	18.6	21.6	25.2	28.5
		MCWBR	14.4	12.1	9.5	7.2	4.9	3.9	3.9	4.6	5.0	6.6	10.0	13.4
	5% WB	MCDBR	27.9	28.4	25.1	21.3	14.5	12.3	12.6	14.5	15.4	16.0	18.0	22.7
	MCWBR	13.4	11.3	8.5	6.6	4.2	3.7	3.8	4.5	4.6	5.1	7.5	11.2	
Clear-Sky Solar Irradiance	taub	0.337	0.387	0.613	0.607	0.472	0.424	0.418	0.451	0.442	0.418	0.361	0.340	
	taud	2.403	2.262	1.653	1.736	2.187	2.388	2.393	2.274	2.275	2.306	2.398	2.421	
	Ebn at Noon	294	284	226	232	264	275	276	268	269	271	283	288	
	Edh at Noon	34	42	80	75	48	38	38	43	42	39	34	32	
All-Sky Solar Radiation	RadAvg	1387	1665	1719	1727	1559	1331	1217	1332	1342	1284	1335	1237	
	RadStd	102	122	127	135	166	130	82	103	94	141	103	97	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only			+0.50	+0.61	N/A	+0.92	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)			+0.50	N/A	N/A	+0.76	N/A	N/A	-61	+205	+146

Nomenclature: See separate page